

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 90

THE QUINN SELF-OSCILLATING ECCENTRIC WEIGHT CENTER-OF-GRAVITY ALL-TERRAIN MOTORS

When the ground goes flat, carry your own hill
an offset weight rolls the ball from inside



a ground machine — the vet confused it, then corrected itself

CENTER-OF-GRAVITY DRIVE · LINKAGE-FREE

SHIFT THE WEIGHT — ROLL THE WORLD

Ground-contact propulsion — v1.0 in force

GP-IM-090

Revision v1.0 — 22 August 2026

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VETTED — 4-AI WEB — BATCH 16 — STANDS

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 90

PHASE 90: THE QUINN SELF-OSCILLATING ECCENTRIC WEIGHT CENTER-OF-GRAVITY ALL-TERRAIN MOTORS — STATUS: CURRENT. When the ground goes flat, gravity quits — so the ball carries its own hill inside it. Phase 90 suspends a sealed spherical kinetic drive pod at the exact geometric center of the Phase 89 geodesic frame: a heavy offset counterweight on a low-power, high-torque motor (Phase 71 sodium-ion bank), continuously shifting the assembly's center of gravity so the ball rolls over dead-flat valleys, dunes, and uphill grades. No wheels, no steering rods, no exposed joints — the unibody rotation is the drivetrain. This is a GROUND machine, and the record is exact about it: internal eccentric masses cannot translate a free-floating vehicle — momentum cancels over the complete cycle — but against terrain, vibration and weight-shift do useful work. The vet confused this phase with the dumped Phase 37 mass-shifting energy system, was asked to its face, and corrected itself: Phase 90 STANDS.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-090, v1.0 — 22 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the eccentric-drive law as seated (contents line, the two design bodies — verbatim) — §2 why it works (ground contact versus free space — graded) — §3 the system in the pod — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 90: **Self-Oscillating Eccentric Weight Center-of-Gravity All-Terrain Motors.** Propels heavy all-terrain cargo sleds across rugged planetary surfaces by sequentially shifting off-center, eccentric internal weights, creating cyclic center-of-gravity lurches to drive traction across loose gravel or slick ice.'

THE MASTHEAD LINE (verbatim): 'Workshop Toy Analog: Motorized Self-Rolling Children's Pet Balls & Internal Pendulum Gyros [extracted verbatim from the combined masthead line]'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Active Self-Propulsion Across Flat Extraterrestrial Terrain Baselines [extracted verbatim from the combined masthead line]'

ECCENTRIC CENTER-OF-GRAVITY DRIVES (VERBATIM)

* Core Assembly: Suspends a sealed, spherical kinetic drive pod in the exact geometric center of the Phase 89 expandable scissor-joint geodesic frame. * The Asymmetrical Pendulum Cam: Integrates a heavy, offset internal counterweight driven by a low-power, high-torque motor fueled by Phase 71 Sodium-Ion batteries. * Continuous Center Shift: Continuous rotation of the internal mass shifts the drone's center of gravity permanently outside its structural footprint, forcing the expanded sphere to perpetually fall forward into an active, high-velocity rolling velocity vector.

INDEPENDENT ALL-TERRAIN PROPULSION (VERBATIM)

* Flat Plain Traversal Mode: Overrides gravitational slope deceleration thresholds. The internal oscillating engine engages automatically upon transitioning onto dead-flat valleys, sand dunes, or uphill planetary grades. * Linkage-Free Kinematics: Eliminates fragile external wheels, steering rods, and exposed joints. The unibody rotation of the outer S-bend sugar-scoop shell provides enclosed mechanical propulsion, permanently insulating the drive from abrasive lunar dust sandblasting or gear-tooth shear lockups.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE FREE-SPACE BAR, AFFIRMED AND QUOTED: internal eccentric masses cannot translate a free-floating vehicle — momentum cancels over the complete cycle. The audit's own Register Entry 2 and the vet agree word for word. Phase 90 never claimed otherwise.

THE GROUND-CONTACT CASE, AFFIRMED — THE MACHINE'S TRUE PHYSICS: against terrain, a shifting center of gravity is propulsion — the ball falls toward its own offset weight, continuously, and the ground closes the momentum loop. The vet said it itself: 'for a ground-contact machine, vibration can absolutely interact with the terrain and produce useful motion. That's a legitimate mechanical class.' Self-rolling toys prove the class on every kitchen floor.

THE LINKAGE-FREE CHASSIS, AFFIRMED: no wheels, steering rods, or exposed joints means nothing to jam with dust, freeze, or shear — the failure modes of conventional rovers are deleted by omission, not managed by engineering.

THE DRIVE BUDGET, AFFIRMED AS SIZED: a low-power high-torque motor turning an offset mass is a small, steady draw — sized to the Phase 71 sodium-ion bank in the law text.

§3 — THE SYSTEM IN THE POD

- THE POD: sealed spherical kinetic drive suspended at the geometric center of the Phase 89 frame.
- THE WEIGHT: heavy offset counterweight — the asymmetrical pendulum cam — on a low-power, high-torque motor.
- THE BANK: Phase 71 sodium-ion cells feeding the drive; steady draw, no surge class.
- THE MODES: passive gravity tumble on slopes; the oscillating engine engages automatically on dead-flat valleys, dunes, and uphill grades.
- THE SHELL: unibody rotation — no wheels, rods, or exposed joints anywhere in the propulsion path.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE CARRY-YOUR-OWN-HILL DOCTRINE (seated with the pod): when the terrain stops helping, bring the slope with you — the offset weight is a portable grade.
- THE GROUND-VS-SPACE DOCTRINE (the batch-16 lesson, both audits agreed): the same internal mass that is fraudulent in free space is legitimate on the ground — the difference is the contact, and the record now says so in both audits' words.
- THE DELETE-THE-JOINTS DOCTRINE (seated with the chassis): what is not there cannot jam — the unibody is reliability by omission.
- THE TOY-TO-TOOL DOCTRINE (the masthead analog, verbatim): the motorized self-rolling pet ball and the internal pendulum gyro — the class was always real; the phase grows it up.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the pod: sealed spherical drive, offset mass on the high-torque motor; balance map logged across the full rotation.
- STEP 2 — Mate the frame: pod suspended at the geometric center of the Phase 89 geodesic shell; concentricity gauged.
- STEP 3 — Wire the bank: Phase 71 sodium-ion feed; draw profile benched across duty cycles.
- STEP 4 — Bench the traction: roll-out on flat, sand, and grade; traction versus surface class logged — the number this machine lives or dies by.
- STEP 5 — Prove the mode switch: automatic engagement on slope-to-flat transitions; hysteresis

tuned so the drive never hunts.

- STEP 6 — Publish the ledger: traction curves, draw, duty cycle — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 89 — the roll-ball frame (GP-IM-089): the shell this motor drives; 89 owns the slopes, 90 owns the flats.
- PHASE 71 — the sodium-ion matrix (GP-IM-071): the drive bank.
- PHASE 37 — the Mayernick ruling (GP-IM-037): the killed mass-shifting ENERGY system this phase is not — the distinction both audits now enforce.
- PHASE 95 — the prism-lock feet (GP-IM-095): anchoring the harvester's work sites.

§8 — DO-NOT-BUILD

- NEVER sell it as free-space propulsion — internal masses cannot translate a free-floater; the ground is the machine's other half (Register Entry 2, both audits).
- NEVER resurrect the Phase 37 conflation — the dumped flywheel energy system and this ground drive are different machines; the record is exact.
- NO external drivetrain grafts — wheels, rods, and exposed joints are the failure modes this phase deleted.
- NEVER run the drive on the tumble slopes — gravity is free; the motor is for where gravity quits.

§9 — OWED

THE TRACTION BENCH, OWED: roll traction versus surface class (regolith, sand, sintered ice, plate rock); counterweight duty cycle and draw profile against the Phase 71 bank; mode-switch hysteresis figures.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-090, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 22 August 2026, third of the 20-chapter 'clear the 100 mark' shift ('do another 20 while i away so we clear the 100 mark for the day when i get back'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i

will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564 and 566): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 90